

CBSE Pattern Question Bank

Chapter 1: Natural Resources and Their Use

Class 8 | Exploring Society: India and Beyond

SECTION A — Very Short Answer Questions (1 Mark each)

Q1. What is the term used for materials and substances that occur in Nature and are valuable to humans?

Ans. Natural Resources.

Q2. Name any two non-renewable resources mentioned in the chapter.

Ans. Coal and Petroleum (also iron, copper, gold are acceptable).

Q3. What does 'exploitation' mean in the context of natural resources?

Ans. In this context, exploitation means extraction, utilisation, and consumption of natural resources — not harming them.

Q4. What is biodiversity loss?

Ans. Biodiversity loss is the decline in the variety of life on Earth.

Q5. In which year did Sikkim become a 100% organic state?

Ans. In 2016, Sikkim became a 100 per cent organic state with all its farmland certified organic.

Q6. Name the ancient Indian botanical science that focuses on the study and care of plants.

Ans. Vriksháyurveda — an ancient Indian botanical science dealing with the study and care of plants and trees.

Q7. What does the term 'arghyam' mean?

Ans. Arghyam means an offering, generally of water, as a mark of respect or gratitude.

Q8. Who launched the International Alliance for Solar Energy (IASE) and in which year?

Ans. India and France launched the IASE in 2015.

Q9. How much oxygen does a mature tree produce per day approximately?

Ans. A mature tree produces approximately 275 litres of oxygen per day.

Q10. What is meant by 'natural resource curse' or 'paradox of plenty'?

Ans. It refers to the phenomenon where regions rich in natural resources experience slower economic growth and development, because they fail to develop industries that convert resources into higher-value products.

SECTION B — Short Answer Questions (2–3 Marks each)

Q11. State three conditions necessary for an entity to be called a natural resource.

Ans. For an entity to be called a resource it must be:

(i) Technologically accessible — we must have the technology to extract or use it.

(ii) Economically feasible — the cost of extraction must be justifiable.

(iii) Culturally acceptable — its use must not violate cultural or social norms (e.g., cutting trees in sacred groves is unacceptable).

Q12. Differentiate between restoration and regeneration.

Ans. Restoration is the process of returning something to its original healthy state after it has been degraded or damaged (e.g., a forest recovering after wildfire). Regeneration goes beyond restoration — it is Nature's ability to create new life and conditions for thriving (e.g., a fallen tree decomposing, enriching the soil, enabling new plants to grow).

Q13. What is the 'natural resource curse'? How has India generally avoided it?

Ans. The natural resource curse (or paradox of plenty) means that having abundant natural resources does not automatically make a country rich, because economies often fail to develop industries that convert resources into higher-value products. India has generally avoided this curse by investing in the development of industries to process and use its natural resources to meet its growing needs.

Q14. Explain why solar energy and timber are called renewable resources. What one condition must be met?

Ans. Solar energy is continuously available as sunlight replenishes daily. Timber from forests is renewable because forests can grow back naturally. However, the condition is that the natural rhythm of restoration and regeneration must not be disturbed. For example, if we harvest timber faster than forests can grow trees, the forest will eventually be depleted.

Q15. Why are fossil fuels classified as non-renewable resources? How long are India's coal reserves estimated to last?

Ans. Fossil fuels like coal and petroleum are classified as non-renewable because they were formed over millions of years and cannot be replenished at the rate we use them. It has been estimated that India's coal reserves may last another 50 years, while the demand for electricity continues to increase.

Q16. What are ecosystem functions and ecosystem services? Give one example of each.

Ans. Ecosystem functions are Nature's inherent ways of working — for example, a forest filtering water, preventing soil erosion, and providing habitat for animals. Ecosystem services are the benefits humans receive from these functions — for example, clean water, protected farmland, and pollinated crops that humans receive because of the forest.

Q17. What problems did the overextraction of groundwater cause in Punjab?

Ans. In Punjab:

- (i) The groundwater level became inaccessible till depths of about 30 metres in large parts.
- (ii) Chemicals from pesticides and fertilisers dissolved in groundwater, causing serious health hazards.
- (iii) Almost 80% of Punjab's area was classified as 'overexploited'.
- (iv) The rate of extraction was far greater than the rate of natural restoration and rejuvenation of groundwater.

Q18. State any three categories of natural resources based on their uses.

Ans.

- (i) Resources essential for life — air, water, soil/food. We cannot make these ourselves.
- (ii) Resources for materials — wood, marble, coal, gold; used to create physical objects for utility or beauty.
- (iii) Resources for energy — coal, water, petroleum, natural gas, sunlight, wind; used to generate electricity and power transportation and production processes.

Q19. How does the uneven distribution of natural resources affect human life?

Ans. Uneven distribution shapes:

- (i) Human settlements — people settle near resource-rich areas.
- (ii) Trade patterns — nations trade what they have for what they lack.

(iii) Employment — industries near resources create local jobs and townships.

(iv) International relations and conflicts — many wars have been fought to gain control over natural resources.

(v) Tensions over sharing — e.g., sharing of Kaveri River water among Karnataka, Tamil Nadu, Kerala and Puducherry.

Q20. What is Vriksháyurveda? Name two sustainable practices it promotes.

Ans. Vriksháyurveda is an ancient Indian botanical science focused on the study and care of plants and trees. The term comes from Sanskrit — 'vriksha' meaning tree and 'ayurveda' meaning science of life. It was formalised around the 10th century CE in texts like Surapala's Vriksháyurveda.

Two sustainable practices it promotes:

(i) Crop rotation and mixed cropping to maintain soil health.

(ii) Proper ploughing methods to retain soil moisture and support soil organisms like fungi, bacteria, and earthworms.

SECTION C — Long Answer Questions (5 Marks each)

Q21. Describe in detail the groundwater crisis in Punjab. What caused it and what are its consequences?

Ans. Punjab, known as the home of India's Green Revolution, now faces a severe groundwater crisis:

Causes:

(i) In the 1960s, farmers shifted to high-yielding varieties of wheat and paddy that required more water than traditional seeds.

(ii) The supply of free power led to over-pumping of groundwater.

(iii) Use of chemical pesticides and fertilisers became widespread.

Consequences:

(i) Groundwater became inaccessible till depths of about 30 metres across large parts of Punjab.

(ii) Chemicals dissolved into groundwater causing health hazards to people.

(iii) Almost 80% of Punjab's area was classified as 'overexploited'.

(iv) Food security was ensured in the short term but at the cost of long-term sustainability.

This shows that exploiting nature beyond its capacity to regenerate leads to serious long-term consequences for both people and the environment.

Q22. What is meant by stewardship of natural resources? Explain with examples how we can be responsible stewards of both renewable and non-renewable resources.

Ans. Stewardship means using natural resources in a responsible and wise manner so that they can be restored, regenerated, and sustained for future generations.

For Renewable Resources:

(i) We must not harvest resources faster than they can regenerate — e.g., timber should not be cut faster than forests can grow.

(ii) Traditional water harvesting, rejuvenation of ponds and tanks, and reduction of wasteful water consumption help maintain groundwater levels.

(iii) Using natural fertilisers (cow dung), mulching, multi-cropping, and organic farming (as done in Sikkim) helps maintain soil health.

(iv) Communities should regulate fishing during spawning seasons to maintain fish populations.

For Non-Renewable Resources:

(i) We should use them judiciously so they last long enough for humanity to develop sustainable alternatives.

(ii) Switching to renewable energy sources — solar, wind, water — reduces dependence on coal and petroleum.

(iii) India's Bhadla Solar Park in Rajasthan and participation in IASE are examples of this transition.

The Bhagavad Gita's concept of lokasangraha — acting for the wellbeing of all — guides us toward this responsibility.

Q23. Explain the concept of Nature's cycle of restoration and regeneration with two examples. What happens when this cycle is disturbed?

Ans. Nature works in cycles where there is no waste and everything is renewed.

Example 1 — Forest Cycle: A tree falls in the forest. Bacteria, fungi, and insects decompose it. It becomes part of the soil, enriching it. New trees grow from seeds and the cycle repeats.

Example 2 — River Cycle: Rivers are fed by rainfall and melting glaciers. Water evaporates, forms clouds, rains again — a continuous cycle that replenishes river water.

When the Cycle is Disturbed:

(i) Groundwater: When extraction rate exceeds replenishment rate (as in Punjab), groundwater becomes unavailable. Cities risk running out of groundwater.

(ii) Glaciers: Fossil fuel-driven industrialisation and deforestation have led to rising temperatures. Himalayan glaciers are melting faster than precipitation can replace them, threatening water security of plains populations.

(iii) Industrial waste discharged into rivers disrupts Nature's cycle — rivers become polluted and cannot support life.

(iv) Overfishing disrupts ocean ecosystems — tuna populations declined drastically when commercial fishing ignored spawning seasons.

Conclusion: Irresponsible human actions leading to pollution, biodiversity loss, and climate change are the main reasons for disruption of Nature's restorative cycles.

Q24. 'Having abundant natural resources does not guarantee economic prosperity.' Explain this statement and describe how India has addressed this challenge.

Ans. This statement refers to the 'natural resource curse' or 'paradox of plenty' — the phenomenon where resource-rich regions experience slower economic growth.

Why this happens:

(i) Countries with abundant resources may not develop industries to convert raw materials into higher-value products.

(ii) They may rely only on selling raw resources, which gives lower income compared to manufactured goods.

(iii) Poor governance and lack of strategic planning mean resources become temporary windfalls rather than lasting benefits.

How India has addressed this:

(i) India has generally avoided the resource curse by investing in the development of industries that process natural resources.

- (ii) India has developed its manufacturing and technology sectors alongside resource extraction.
- (iii) India has taken leadership in renewable energy — co-founding IASE with France in 2015, and building one of the world's largest solar parks (Bhadla Solar Park) in Rajasthan.
- (iv) India's approach shows that human knowledge, good governance, and strategic planning are key to turning natural resources into lasting economic benefits.

SECTION D — Source-Based / Case Study Questions (4 Marks each)

Q25. Read the following passage and answer the questions: "Many farmers in our country extract groundwater for irrigation purposes. In most states, the extraction rate is greater than the rate at which the water table is replenished. Over time, this deficit builds up..."

(a) Why is the rate of extraction greater than replenishment in Punjab?

Ans (a). Because high-yielding varieties of wheat and paddy require more water; free power supply led to over-pumping; and modern farming needed heavy chemical use — all together causing groundwater to be extracted much faster than it can be naturally replenished.

(b) Name any two consequences of overextraction of groundwater.

Ans (b). (i) Groundwater became inaccessible till about 30 metres depth. (ii) Chemicals from pesticides dissolved in groundwater causing health hazards.

(c) Suggest two ways to address this problem.

Ans (c). (i) Traditional water harvesting and rejuvenation of ponds and tanks. (ii) Cutting down wasteful water consumption and processing and reusing water.

(d) What does this case teach us about using renewable resources?

Ans (d). Even renewable resources like groundwater can become non-renewable if extracted faster than they can be replenished. Sustainable use and mindful management are essential.

Q26. Read the following and answer the questions: "In 2016, Sikkim became a 100 per cent organic state... local biodiversity flourished, with beneficial insects and birds returning; tourism increased... and farmers' incomes grew by 20 per cent on average."

(a) What problems did Pema's family face before switching to organic farming?

Ans (a). Pema's family faced declining yields and mounting debts from expensive chemical inputs.

(b) What natural methods did the family use after switching to organic farming?

Ans (b). The family switched to compost, prepared natural pest repellents using neem and garlic, and started growing multiple crops across the year.

(c) What were the environmental benefits seen in Sikkim after going organic?

Ans (c). Local biodiversity flourished, with beneficial insects and birds returning to the region.

(d) Why is Sikkim considered a global model?

Ans (d). Sikkim demonstrated that an entire region can successfully transition to sustainable agricultural practices while improving both ecological outcomes (biodiversity, soil health) and economic outcomes (20% income growth, increased tourism).

SECTION E — Higher Order Thinking Skills (HOTS) Questions (3 Marks each)

Q27. 'Nature does not pay attention to political boundaries.' What conflicts can arise from this and how can they be resolved?

Ans. Rivers, groundwater, and mineral resources do not follow state or national borders. This leads to disputes over sharing — e.g., Kaveri River water is shared among Karnataka, Tamil Nadu, Kerala, and Puducherry. Such conflicts require careful negotiations, agreements, and deft management to maintain peace. Internationally, reaching agreements between countries is even harder. Sharing of the Brahmaputra (Yarlung Tsangpo) river is one such international example. Resolution requires diplomatic cooperation, fair treaties, and international water-sharing agreements.

Q28. How can a renewable resource become non-renewable? Give a real example from the chapter.

Ans. A renewable resource becomes non-renewable when it is used faster than it can regenerate. Example from the chapter: Groundwater in Punjab is a renewable resource in principle — it is replenished through rainfall. But because extraction rates were far greater than replenishment rates, about 80% of Punjab is now 'overexploited' and groundwater is inaccessible. Another example: Himalayan glaciers — a combination of deforestation and fossil fuel use has caused temperatures to rise, melting glaciers faster than precipitation can replace them, threatening the 'water tower' for plains populations.

Q29. Do you think economic development and environmental sustainability can go together? Use examples from the chapter to support your answer.

Ans. Yes, they can go together, as shown by the chapter's examples:

- (i) Sikkim's organic farming transition improved both ecology (biodiversity, soil health) and economy (20% income rise, tourism growth).
- (ii) India's International Solar Alliance and Bhadla Solar Park show that investing in renewable energy creates both environmental and economic opportunities.
- (iii) Traditional materials like mud and stone combined with modern techniques create sustainable buildings that also provide local employment.

However, balance requires human knowledge, good governance, and strategic planning. Without these, development can damage the environment as seen in Punjab's groundwater crisis.

Note: All answers are based strictly on Chapter 1 content of 'Exploring Society: India and Beyond | Grade 8 Part 1' (NCERT, Reprint 2026-27).